# METHODOLOGY

## Red Tape to Recovery: Tracking Altadena Rebuilding After the Eaton Fire

April 2026

Over 24 days in January 2025, The Eaton Fire rampaged through Altadena burning through 14,000 acres, destroying thousands of properties, and resulting in 19 deaths. A year later, data and personal stories show that only a few Altadenans and Pasadenans have returned home. This research project, developed by Catalyst California in partnership with Dena Rise UP!, provides crucial, accurate data on how different areas and properties are moving toward rebuilding. We analyzed the devastation in West and East Altadena to understand the housing lost, the uphill battle to recovery, and the threats to residents returning home to help support a just and inclusive reconstruction.

While the Eaton Fire affected both the Altadena and Pasadena communities, we focused our analysis on Altadena given the fire's disproportionate impact on this community and its legacy of Black homeownership and wealth. This also allowed us to focus on understanding progress toward rebuilding for unincorporated L.A. County. We compared the devastation in West and East Altadena by splitting Altadena's unincorporated community boundaries by West and East of Lake Avenue. By comparing West and East Altadena, we can better understand the fire's impact on these neighborhoods and their specific challenges to rebuilding.

Our analysis focused on residential properties that were significantly damaged by the Eaton Fire. We relied on five main data sources to identify and track characteristics of significantly damaged properties.

**Damage Data**
California Department of Forestry & Fire Protection. (2025). *CAL FIRE damage inspection (DINS) data*. California Natural Resources Agency GIS. https://gis.data.cnra.ca.gov/datasets/CALFIRE-Forestry::cal-fire-damage-inspection-dins-data/.

**Permit Data**
L.A. County. (2025, December 10). *Electronic permitting & inspections portal (EPIC-LA)*. L.A. County Department of Public Works & Department of Planning. https://epicla.lacounty.gov/energov_prod/SelfService/#/search.

**Residential Property Characteristics**
L.A. County Assessor. (2025). *Secured basic file abstract.* https://assessor.lacounty.gov/real-estate-toolkit/data-sales.

**Property Sales Data**
Altadena Not for Sale. (March 30, 2026). *Properties sold after the Eaton Fire data*. Privately acquired data.

**Fire Hazard Severity Zones**
California Office of State Fire Marshall. (2024-25). *Fire hazard severity zones.* https://osfm.fire.ca.gov/what-we-do/community-wildfire-preparedness-and-mitigation/fire-hazard-severity-zones.

**Soil Lead Contamination Data**
L.A. County Department of Public Health (2025, September 5). *Community soil sampling and human health screening report*. Roux Associates, Inc. https://storymaps.arcgis.com/stories/667412ef37ee4392bddb5c90da3480f1?cover=false.

**Commented [MK1]:** Do we want to add Altadena Not For Sales here?

Example citation:
Altadena Not For Sales, Properties Sold Data

Altadena Not for Sales. (2026). Properties sold after the Eaton Fire data. Retrieved 00/00/0000.

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# Damage Types

We utilized data from CAL FIRE's Damage Inspection database to identify residential properties lost in the fire. CAL FIRE provides every structure on a property with its own damage assessment. Structures are identified as "No Damage", "Affected (1-9% destroyed)", "Minor Damage (10-25% destroyed)", "Major Damage (26-50% destroyed)", or “Destroyed (>50% destroyed)”. We matched these structure level records to their corresponding parcel level records to be able to track their property characteristics and permit history. Because one property, or parcel, can have multiple structures on it, e.g., a single-family home and detached garage or a single-family home and accessory dwelling unit (ADU), we had to create a typology to classify parcels as significantly damaged or destroyed when they had multiple structures with different damage assessments.

We matched structures to their property or parcel records using three methods: (1) joining structure points to parcel polygons, (2) joining structures to parcels based on AIN, and (3) joining structures to parcels based on site address. We then classified properties into these categories based on their structures’ damage assessments:

- **No Damage**: Parcels that had no damage recorded on any structure on the property or were inaccessible.
- **Some Damage**: Parcels that included at least one structure that was affected (1-9% destroyed) and/or had minor damage (10-25% destroyed) alone or in combination with structures that had no damage.
- **Significantly Damaged or Destroyed**: Parcels that included at least one structure that had major damage (26-50% destroyed) or destroyed (>50% destroyed) alone or in combination with another damage category.

We focused on the last category of properties “significantly damaged or destroyed” in our dashboard and report to track residential properties that sustained the greatest damage and are most likely in need of pursuing permits to rebuild.

# Rebuild Progress

To track the rebuilding process, we collected fire debris removal data from the U.S. Army Corps of Engineers (USACE) and permit data from L.A. County’s Electronic Permitting and Inspections portal (EPIC-LA). We used data from USACE to identify whether properties had completed fire debris removal and permit data from EPIC-LA to identify properties that removed debris without the USACE and/or had started the permit process. Beyond fire debris removal permits, we specifically tracked CREB and UNC permits related to primary residences (e.g., condos, single-family, and multifamily), ADUs, and other miscellaneous structures like garages, sheds, etc. Only permits with application dates after the start of the Eaton Fire were reviewed and permits with “Void”, “Canceled”, or “Denied” statuses were excluded.

Based on the permit data at the time of collection, we classified each property into one of five rebuild stages. Our rebuild progress numbers may differ from L.A. County’s Permitting Progress Dashboard due to differences in methodology. For instance, we include properties beyond new construction permits to account for significantly damaged properties that can rebuild with repair permits. Our project also does not account for repairs made without permits.

- **Without Permit Applications, Fire Debris Removal Incomplete**: There is no indication, either by USACE or the property owner, that fire debris has been removed. Fire debris removal is required before a permit application can be issued. For instances where property owners opted out or were considered ineligible for USACE fire debris removal support, we reviewed fire debris removal permits from the EPIC-LA portal to determine if this stage was completed. Additionally, if a property had already acquired a CREB/UNC permit, we assumed fire debris removal was completed.
- **Without Permit Applications**: Fire debris has been removed, either by USACE or by the property owner, but the owner has not applied for a building permit. In other words, we found no CREB or UNC permit applications associated with the property in EPIC-LA.
- **With Permit Applications**: A building permit application (CREB/UNC) has been filed and/or issued, but there is no indication that construction has started. This category indicates that the owner has applied for a permit. It can also include permits that have been approved/issued but for which there is no documentation in EPIC-LA to demonstrate that construction has begun.
- **In Construction**: A building permit application has been issued, and at least one inspection has been scheduled, indicating construction has started. We identified construction as started based on workflow status items collected through EPIC-LA. If there was at least one workflow item with a description that included inspection, we counted the property as in construction.
- **Repairs or Rebuild Complete**: All building permits related to permanent, temporary, and miscellaneous structures have been marked “finaled”. This category can include properties that have completed repairs like a new roof or properties that were full rebuilds (e.g., construction of a new house that was destroyed by the fire). If a property only had temporary housing permits “finaled”, we did not include the property in this category, and it remains “In Construction”. Similarly, if a property only had trade (e.g., solar panels, sewers, mechanical), plumbing, or electrical permits “finaled”, we did not count the property as completed. We manually check permit records on properties that move into the repairs or rebuild complete category to validate whether all repairs or rebuild work have likely been completed based on the property’s damage levels.

# Sold Properties

Based on data from the L.A. County Assessor and Altadena Not for Sale, we mapped and analyzed the properties that have been sold since the Eaton Fire. We used February 8th, 2025, as a cut-off for sales occurring after the Eaton Fire. On January 21st, 2025, L.A. County Sheriff's Department allowed residents to return home and news reports indicate that the first property sold around February 8th, 2025. We checked property sales records between January 21st and February 8th, 2025 and determined the first property sold after February 8th, 2025 was also the first property with evidence of being listed after residents returned on January 21st, 2025.

Because there can be a four-to-six-week lag between L.A. County Assessor data and real-time sales, we combined data from the L.A. County Assessor with sales data gathered by the Altadena Not for Sale (ANFS). The coalition is a public group of citizens that track property sales via public and private domain sites, which are often updated in real time. Any properties that are found to be sold in the ANFS data but not captured by the L.A. County Assessor data are added to the sales list. Our published sales data only includes property sales through the date indicated on the dashboard.

# Residential Property Characteristics

To analyze property characteristics before and after the fire, we gathered L.A. County Assessor data from January and November 2025. We focused on residential properties in Altadena, or parcels with a residential use code that were not recorded as vacant land. While we initially included mixed use properties, we found these properties to comprise a very small share of Altadena properties. In January 2025, only 20 properties were mixed use properties with residential units. L.A. County Assessor was the best, most accessible source of housing stock for this project, but also may misrepresent housing density or rental units in Altadena if property rolls are not updated timely with information on ownership or rental units and/or if property owners have built unpermitted ADUs that are not reflected in their property records. This data does not account for those cases hidden density on Altadena properties.

## Residence Type

We distinguished between single-family, multifamily, and condominiums properties to track progress and challenges toward rebuilding for each of these property types. Based on use codes from the L.A. County Assessor's data, we categorized properties into these three residential types. Single-family homes are properties that are made up of one unit or house with the use code prefix "01". Most properties in Altadena prior to the Eaton Fire were single-family homes. Multifamily homes differ from single-family homes in that they have multiple units on the property. Multifamily properties include any property with two or more units or use code prefixes between "02" and "05". In Altadena, we found that most multifamily homes had between two to five units on the property. ADUs can be found on single-family or multifamily properties. A condominium is a building or group of buildings with individually owned units. These include properties with use codes ending in the suffix “E” or “C”. Prior to the fire, there were only 181 condominiums in Altadena.

## Total Square Footage & Units

Properties vary in their total square footage and units. Square footage refers to the total building area or footprint on the property while the number of units refers to the number of individual housing units on the property. We summed the square footage of every building on the property to calculate total residential square footage and summed all units on the property to calculate the total units. Units include renter- and owner-occupied units. Condominiums count as separate units as each condo unit has its own property record. In other words, even though a condominium complex will have 20 units, each individually owned condominium counts as its own unit.

## Ownership Type

We classified the ownership types of each property based on a combination of occupancy and property tax characteristics. We defined ownership types based on the unit types, exemption characteristics, and owner names associated with each property. For properties without recent sales, ownership type is based on parcel data from the L.A. County Assessor’s office. For recent sales where the Assessor’s office has not recorded a property as sold yet, we utilize Altadena Not for Sale’s information on the new owner.

Our ownership types are our closest approximations of each property’s true occupancy and ownership characteristics. In some cases, we may record a property that is owner occupied as something else, such as likely owner occupied, because there are no homeowner exemptions recorded. In other cases, a corporation could own a property but has not reported rental units on the property’s record; therefore, we are unable to classify it as renter occupied.

- **Owner occupied, homeowner exemption**: Properties with a homeowner or veteran’s exemption on the property tax bill, but with no rental units listed on the property record.
- **Likely owner occupied, no exemption**: Properties that are owned by a private individual (and not in a trust, LLC, etc.), but have no exemption type listed on the property tax bill. We assume most of these may be homeowners that have not claimed their homeowner exemption.
- **Renter occupied**: Properties solely with renter-occupied units on the property, and no exemption type listed.
- **Owner and renter occupied**: Properties with a homeowner exemption and rental units listed on the property.
- **Trust owned**: Properties with no rental units or property tax exemptions, e.g., no homeowner exemption, and that are owned in a trust.

**Commented [HK3]:** @Elycia Mulholland Graves based on our last conversation let me know if I should assign you a separate asana task to update this section or if I can include it in Maria's task: https://app.asana.com/1/110506578179264/project/1211010946971619/task/1213974376090108?focus=true

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- **Sold to state**: Properties that are tax delinquent and recorded as being sold to the state. These may include owner- or renter-occupied units.
- **Corporation owned**: Properties with no rental units or property tax exemptions and that are owned in a LLC, LP, corporation, or investment company.
- **Government owned**: Properties that are government owned and have no rental units.
- **Church, charity, or nonprofit owned**: Properties with a church, religious, or welfare tax exemption, or that are owned by churches, social service organizations, community development corporations, or nonprofits, but have no rental units. This category includes nonprofits and other entities that are investing in keeping Altadena owners in their homes through land banking and other financial support.
- **Other ownership**: Properties that are not owner occupied and show no rental units; however, they are also not owned in a trust or corporation. This includes some residential properties owned by water companies and some owned by private investors.

# Fire Hazard Zones

Between April 2024 and March 2025, the California Office of the State Fire Marshall (OSFM) finalized fire hazard severity zones in state responsibility areas and local responsibility areas. Property owners within a high or very high fire hazard severity zone must comply with defensible space requirements around their property, among other regulations. Fire hazard severity zones evaluate "hazard," not "risk". The State Fire Marshall defines "hazard as based on the physical conditions that create a likelihood and expected fire behavior over a 30 to 50-year period without considering mitigation measures such as home hardening, recent wildfire, or fuel reduction efforts". Risk is defined as "the potential damage a fire can do to the area under existing conditions, accounting for any modifications such as fuel reduction projects, defensible space, and ignition resistant building construction." We downloaded the fire hazard severity zone boundaries and matched them to properties in Altadena based on spatial intersect. If a property crossed multiple fire hazard severity zones, we kept its highest severity zone. We distinguished between fire hazard authority areas because a property owner with a home in a very high severity zone within state responsibility may have different requirements than a property in a very high local responsibility area.

# High-Lead Soil Grids

We used data from the L.A. County Department of Public Health and Roux Associates, Inc. to assess lead hazards in soil surfaces across Altadena. Between February and

March 2025, L.A. County Department of Public Health contracted Roux Associates, Inc. to conduct surface soil testing at residential properties near and within the Eaton Fire. Their results confirmed lead as a chemical of potential concern from the wildfire. Lead was the only chemical they found that was (1) above soil screening levels considered protective over a lifetime of exposure, (2) geographically consistent throughout the area, and (3) above background concentrations or ambient levels. Additionally, they found that while lead chemical levels significantly reduced after debris removal, parcels still showed areas with lead concentrations exceeding residential screening levels. Because of these findings, we focused on their soil lead testing results. Importantly, the results are only for soil surfaces and not for indoor contaminants of intact structures.

L.A. County Department of Public Health published average lead geometric means by square grid in Altadena, but they did not publish results by individual property. Their report recommends that properties with soil lead concentrations at or above 80 mg/kg pursue additional precautions and mitigation measures to reduce risk for lead contamination. For properties with children or pregnant people living at the property, they recommend receiving a professional assessment of appropriate mitigation measures. Lastly, for properties with any destroyed structures, they recommend soil impact assessments and management as any part of rebuilding.

Based on these recommendations, we mapped and analyzed testing grids based on whether their geometric mean was above or below 80 mg/kg—the residential soil screening threshold. We classified grids as above or below the threshold, or high lead or not high lead. We then matched properties to these grids by spatial intersect. Importantly, a property's classification is demonstrative of the grid it falls within not its individual property contamination. Individual properties in each grid can vary in their soil lead levels and some properties may have already completed lead remediation.